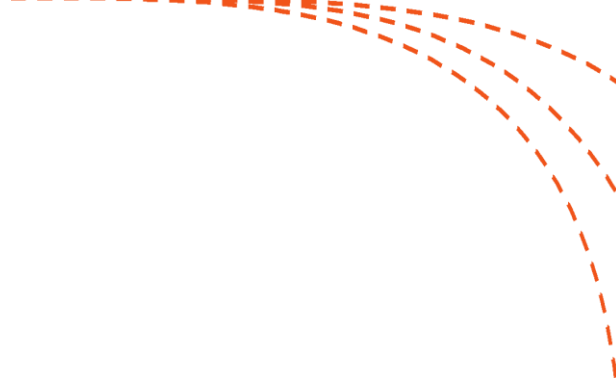




Resilient Mesh Network Testbed



S27-48

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2026/09/22



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Introduction & Objectives

- UAV swarms rely on distributed wireless communications to share telemetry and maintain situational awareness. Since swarm nodes are mobile, the network topology can change as nodes join, leave, or become unavailable.
- The project will develop a ground-based model for studying resilient communications in wireless and ad-hoc networks for UAVs.
- Features 4-6 mobile physical nodes.
- Features a self-healing ad-hoc network protocol.
- Live topology/link-quality visualization tool.
- Final report with quantified resilience metrics.

Problem Statement

- During UAV-swarm missions, including search & rescue, disaster response, and distributed missions. UAV-swarm operators may lose access to timely and reliable communications from the network.
- This can be caused as nodes move, join or leave the network, wireless links degrade, or nodes become damaged and destroyed. These changes can affect coordinated decisions and reduce situational awareness.
- In infrastructure-light environments, mission teams need to understand how the behavior of wireless nodes will be affected when changes to the network and connectivity are affected by mobility and node loss.

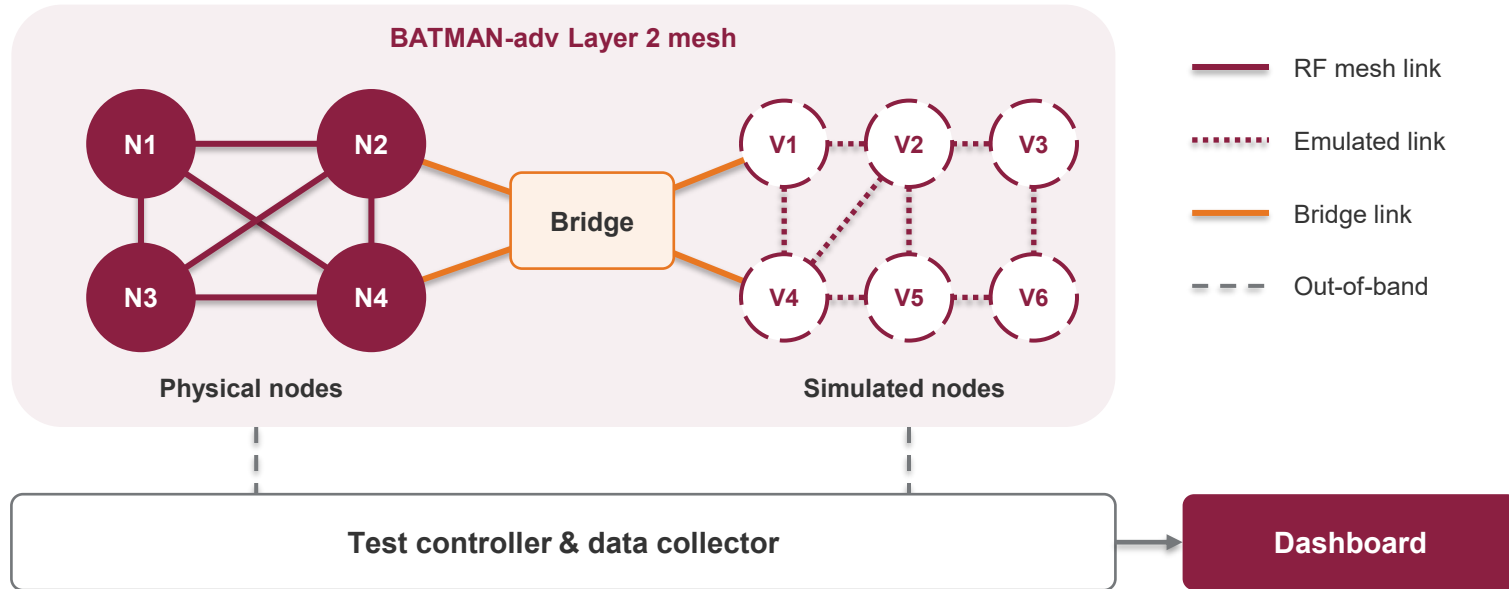
Problem Impact

- Mission-critical data can be interrupted when mobile nodes are lost, and links degrade.
- Communication failures can reduce situational awareness and disrupt coordination and decision making.
- UAV swarms operating in disaster response may lack reliable infrastructure.
- Understanding how networks degrade, recover, or fail helps to design more resilient systems.
- A controlled testbed can provide repeatable and measurable metrics without actual UAV flight.

Proposed Solution

- Our Project
 - 4 physical nodes: SBC (Raspberry Pi target), swappable batteries, optional USB Wi-Fi adapter
 - Up to 6 simulated nodes joined to the physical mesh through emulated links
 - BATMAN-adv Layer 2 mesh routing
 - Test controller that injects failures and timestamps each event
 - Separate out-of-band channel for telemetry and control
 - Web dashboard with live topology and link information
- What makes your solutions innovative?
 - Hybrid physical/simulated mesh: real radio behavior beyond what 4 boards can show
 - Failure validation that separates true protocol failures from test-setup errors

Proposed Solution: System Architecture



Technical Requirements & Constraints

- High-level requirements
 - Ad-hoc, multi-hop mesh, and no central router
 - 4 physical and up to 6 virtual nodes in a mesh
 - Measure reconvergence after node loss
 - Validate protocol failure vs setup error
 - Remote dashboard with live topology
 - 15-20 min battery life
- High-level constraints
 - Separate bands for mesh and out-of-band radio
 - FCC part 15 – only unlicensed bands

Preliminary Information Requirements

- What Raspberry Pi product's onboard radio can hold a stable IBSS link?
 - Bench test with a two-node link, dev forums, and a USB Wi-Fi adapter as a fallback
- How can we join the simulated nodes with the physical mesh?
 - Published hybrid-testbed papers, and candidate tools such as Linux network namespaces
- Which version of the BATMAN protocol is best for our use case?
 - open-mesh.org, the kernel documentation, and side-by-side testing with `batctl`

Preliminary Resource Planning

- What resources do you need?
 - Webhost for remote dashboard.
 - Physical hardware for the non-simulated nodes. (Raspberry Pis, USB Wireless adapters, etc.)
 - Hardware to “sniff” wireless traffic for testing/validation.
 - Possible roof access.
 - Subject matter expert for mesh and Linux networking.
- Where are you going to get them?
 - Contact mentor for access to VT networking resources and for roof access.
 - Submit purchase requests after sufficient research and design of the hardware.
- Is there anything critical?
 - Acquiring the necessary hardware.
 - Webhosting for remote dashboard.

Conclusion & Next Steps

- Design proposal summary
 - Four physical nodes with SBC and swappable batteries.
 - Six simulated nodes with emulated links.
 - BATMAN-adv Layer 2 mesh protocol.
 - Out-of-band control channel for telemetry data.
 - Live web dashboard for topology monitoring.
- Outline of our next steps
 - Build out the initial design test framework on paper.
 - Validate the mesh capabilities for our SBC.
 - Design how the virtual nodes will operate.